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Digitization Platform for the Built Environment

Secured | Scalable | Tailored

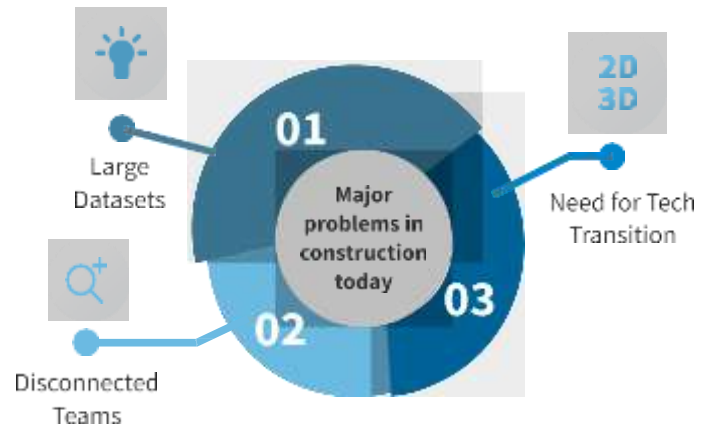


CUBE

Connected Unified Built Environment

DIGITIZATION IS INEVITABLE!

- The need of the hour is Digitization of the construction industry
- Digitization can aid 25% Of Central Govt. Projects which are delayed having a cost overrun of 3 Lakh Crores+
- There are various disconnected tools in the market leading to interoperability issues. It is essential to move to a connected platform for all stages of construction
- With the advent of BIM, Lean Construction, Digital Twin and related tools, there is an opportunity to digitize conventional processes
- Govt. mandates across the globe have mandated digitization in the Built Environment, not just to construction but also operational buildings

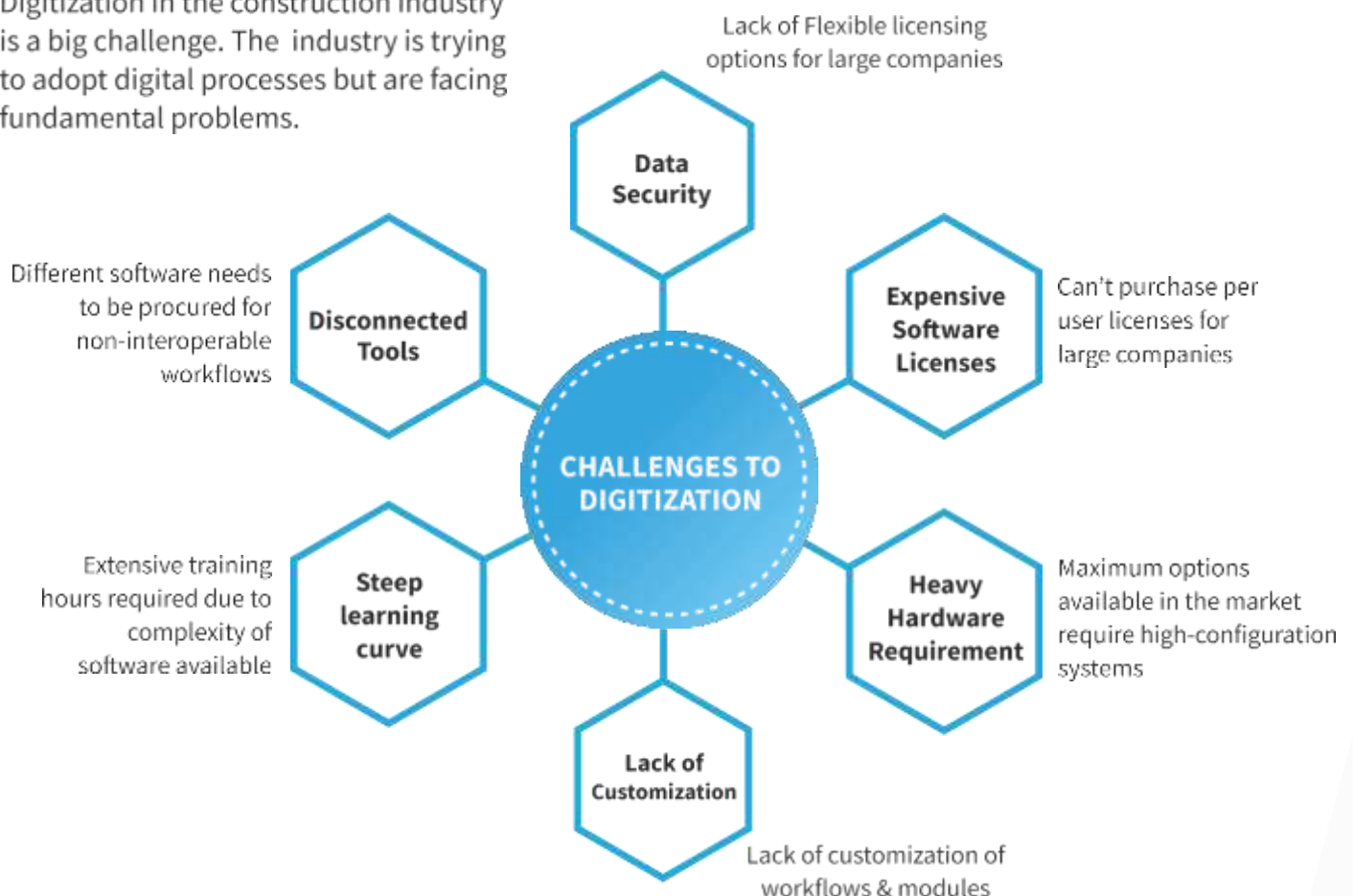


“Project Management Information System (PMIS) is recommended to be used by Government authorities overseeing construction projects to provide real time project updates”

-NITI Aayog Taskforce

DIGITAL TRANSFORMATION CHALLENGES

Digitization in the construction industry is a big challenge. The industry is trying to adopt digital processes but are facing fundamental problems.



OUR SOLUTION

Connected Unified Built Environment (CUBE)



CAD & BIM
VIEWERS



DOCUMENT
MANAGEMENT



DESIGN
COORDINATION



ESTIMATIONS
& COSTS



PROJECT
TRACKING



SITE
CORRESPONDENCE



INTEGRATIONS
WITH ERP



ASSET & DIGITAL
TWINs



“Organizations using Connected Data Environment achieved cost savings of upto 3% of the total project cost”

-Turner & Townsend

INTEGRATED CONSTRUCTION TECHNOLOGY

An end to end integrated tool to manage all stages and types of construction data.



2D Drawings

Manage 2D Documentation
Workflows on the cloud



5D Quantities & Budgets

Integration of 4D models with cost and
quantity-based information



3D Models

View and Review 3D BIM
Models on the Cloud



6D Operations & Maintenance

Integration of BIM models with facility &
asset operation & maintenance data



4D Planning

Integration of 3D models with planned
schedules to create visual simulations

MODULES & FUNCTIONALITIES

CAD & BIM Viewers

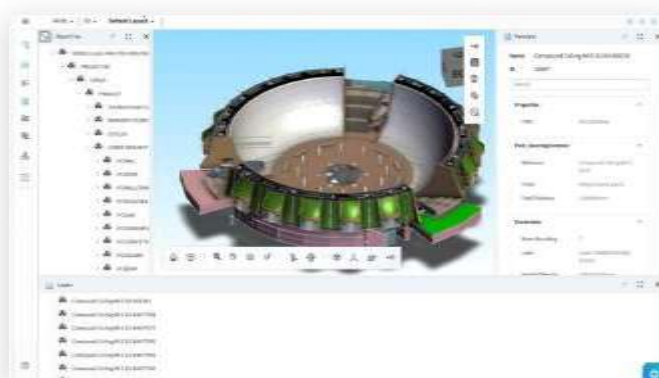
View 2D CAD and 3D BIM data directly from your browser without the need of installing authoring software.



- CUBE offers a modern and responsive Viewer, which is critical to most engineering workflows.
- CAD & BIM Enabled – The entire system is compatible with leading BIM and CAD enabling you to view models directly without the need of additional software licenses or expensive hardware
- Stakeholders can Navigate and review models without installing authoring software from their web browser and mark their comments
- Collaborate in 3D and 2D with all project stakeholders in real time
- Advocates use of open standards like IFC while at the same time also supports vendor specific formats like RVT, NWD, SKP, DWG, OBJ, FBX and many more

3D Viewer - Capabilities

- View any model from top or side views or walk through it, in an interactive manner
- Cut Sections and section boxes
- Take measurements
- View model parameters and interact with model data directly
- Append different models from different stakeholders to create one federated model



2D Viewer - Capabilities

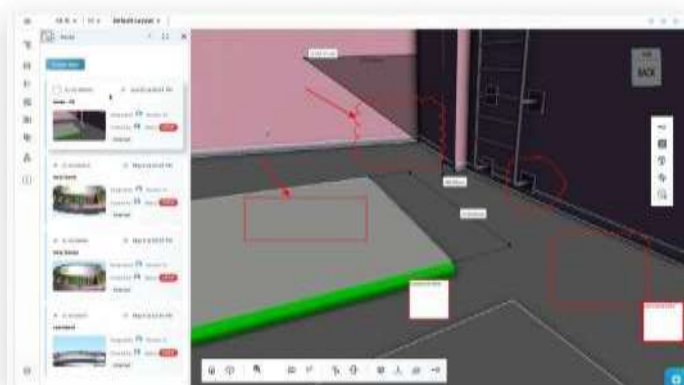
- View CAD files in model space and paper space
- Take linear measurements
- Take drop measurements and automated area measurements
- Point-to-point snapping
- Similar interface to common 2D drafting solutions



MODULES & FUNCTIONALITIES

Design Coordination

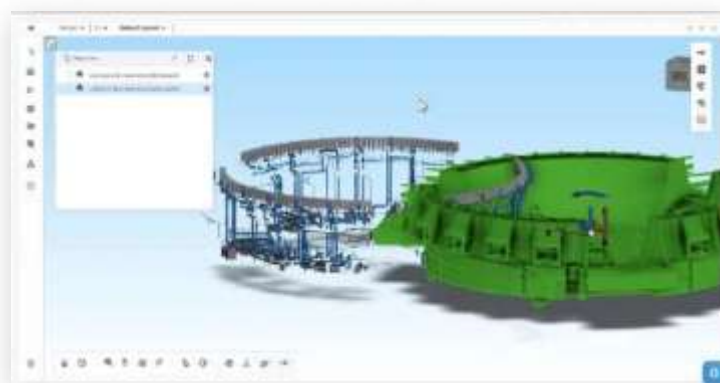
Enhance your digital inclusivity and create coordinated designs by connecting all stakeholders.



- Coordinated Designs & BIM Issues on the cloud.
- Create issues, markups, and comment on 2D drawings and 3D models to communicate changes, discrepancies or disagreements in design and ensure that design development doesn't happen in silos
- Create transmittals to share data across spaces and teams - all while maintaining a detailed log
- Create viewpoints in your design and organize them in sub-folders to make your entire project more manageable

Merge models of different file formats to create a federated BIM model

- Merge files of different formats from your architecture, Structural and MEP consultants directly on CUBE to view a federated model
- Leverage the 3D BIM viewer functionalities to navigate and review the BIM model thoroughly
- Mark issues in these models and associated 2D Drawings to share with respective design consultant for their review



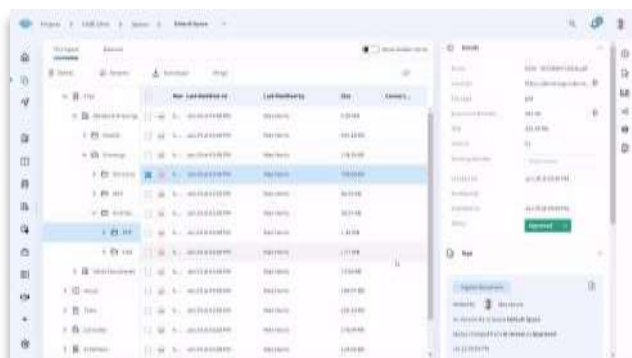
Cloud-based automated Clash Detection Engine

- Run cloud based automated clash detection to review your BIM model.
- Highlight clashes with comments and share transmittals across teams for their review and comments.

MODULES & FUNCTIONALITIES

Document Management

Manage Documents, files and more in an integrated Common Data Environment.



- Easily organize project data and collaborators into purpose built spaces.
- Each Space has an integrated Document management system built-in, which allows users to upload files, in various file formats to the system
- Supports all leading industry - DWG, RVT, IFC, SKP, & many more engineering formats out of the box. In future updates, more formats will be added
- Supports frequently used standard formats like PDF, DOCX, XLSX, PPTX etc that can be viewed & edited without additional licenses or integrations

Site Correspondence

- Users can effortlessly create Digital forms for Site Inspections, Daily Progress Reports, and Health & Safety protocols.
- With customizable templates, diverse field options, and the ability to establish workflows, users get to customize forms for every project.

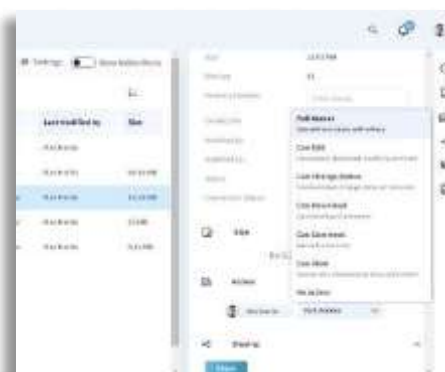


Customizable Approval Workflows

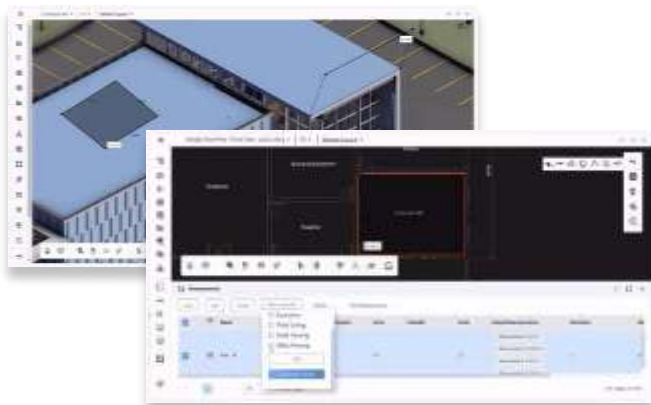
- Make sure your team members don't waste time looking for documents, with easy and comprehensive search functionality
- Store all your items in a central repository equipped with easy and comprehensive search options on metadata and content for documents, drawings, correspondence, emails and more

Advanced Access Control

- Key Document management features like user-control to folders, files and sub-folders, sharing via transmittals, approval workflows, versioning and more.

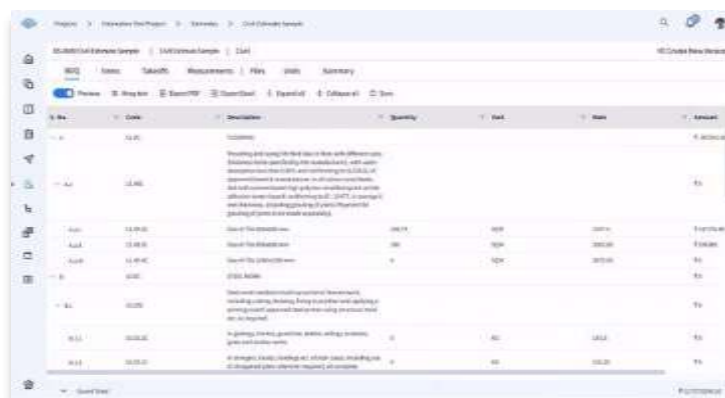


Utilize 2D as well as 3D information to create accurate budgets, BOOs, BOMs, Material Schedules and Estimates.



- Takeoff quantities using various measurement tools
- Measure length, width, height, drop length, area, perimeter, volume, counts, etc. at multiple units of measurement
- Link measurements directly with Cost Databases such as DSR, PAR to create code-compliant estimates
- Created for estimators and surveyors - adaptable and easy to learn

- Advanced filters to manage data and create views for relevant data - teams view only information which is relevant to them
- Automated quantity takeoff with ability to filter, segregate, sort and organize information rapidly
- Integrated approach towards estimation ensures you can use 3D models, 2D drawings and manual measurements to increase accuracy and reduce redundant errors



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MODULES & FUNCTIONALITIES

Progress & Scheduling

The power of VDC and BIM combined to plan and track construction projects.



- Transform your planning schedules on cloud
- The system can interface with Primavera P6 and MS Project for planning, scheduling and monitoring activities and can also generate reports from the schedules, show and track progress including S Curves
- Make edits and changes to imported schedules to extend the utility of planning data
- Create cost quantity and time-quantity s-curves using advanced data visualization features

Powerful 4D/5D simulation engine

- Extend the utility of 3D models and planning data by integrating the two to create comprehensive 4D simulations using our powerful engine
- Enhanced visualization of project plan by linking WBS activities directly with Model and Drawings elements
- Connected 3D to 4D and 5D workflows - estimate, plan and track on the same platform using information across all modules



Reporting through BIM-enabled BI Dashboards

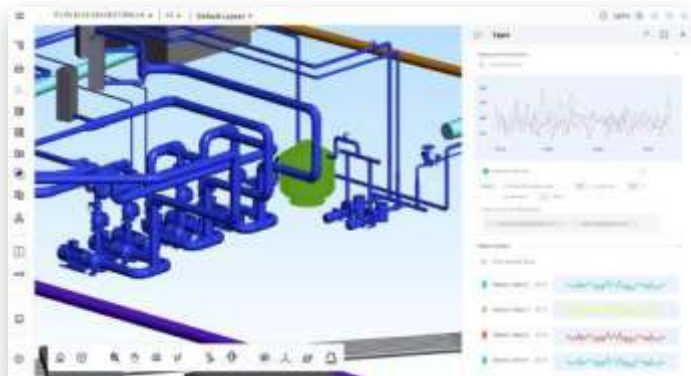
- Leverage BIM data to create comprehensive BI progress reports of your projects
- Schedule and automate your data extraction to create custom BI reports

“Digital Project Management increases visibility, accountability and enables effective decision making through providing real-time and accurate project insights.”

MODULES & FUNCTIONALITIES

Asset & Digital Twins

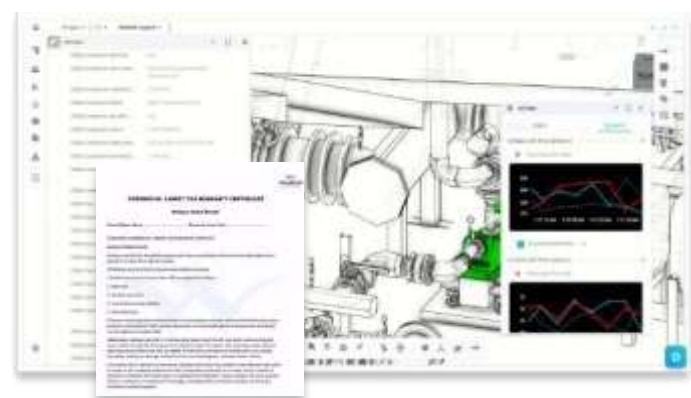
Manage your Operations and Maintenance data through a dynamic digital replica of your built asset on the cloud. "A digital twin for AEC is a dynamic, up-to-date replica of a physical asset or set of assets—whether it's a building, a campus, a city, or a railway—that brings together design, construction, and operational data."



- 24/7 Live tracking of assets through integration of BIM models & IoT Sensors
- Integrate your BIM models to the BMS, CAFM or CMMS systems installed on-site to capture live data on cloud
- Advanced asset tracking and monitoring on the cloud
- Extract asset performance reports for detailed analysis

Asset Management data on cloud

- Track the complete history of assets installed at your facility through a single platform
- Link documents related to your assets like work orders, operation manuals, as-built drawings, equipment specifications etc



Smart Maintenance of Assets

- Keep track of all your assets at your fingertips
- Get notified in real-time when your assets malfunctions or fails.
- The assets in the BIM model changes colours dynamically as the service period nears.

BENEFITS OF HAVING A DIGITAL TWIN

✓ COST SAVINGS

Informed decision-making extends the value and life cycle of assets.

✓ TIME SAVINGS

From day one, owners and operators can begin efficient operation of assets. Never again worry about misplaced documents or complex maintenance documents.

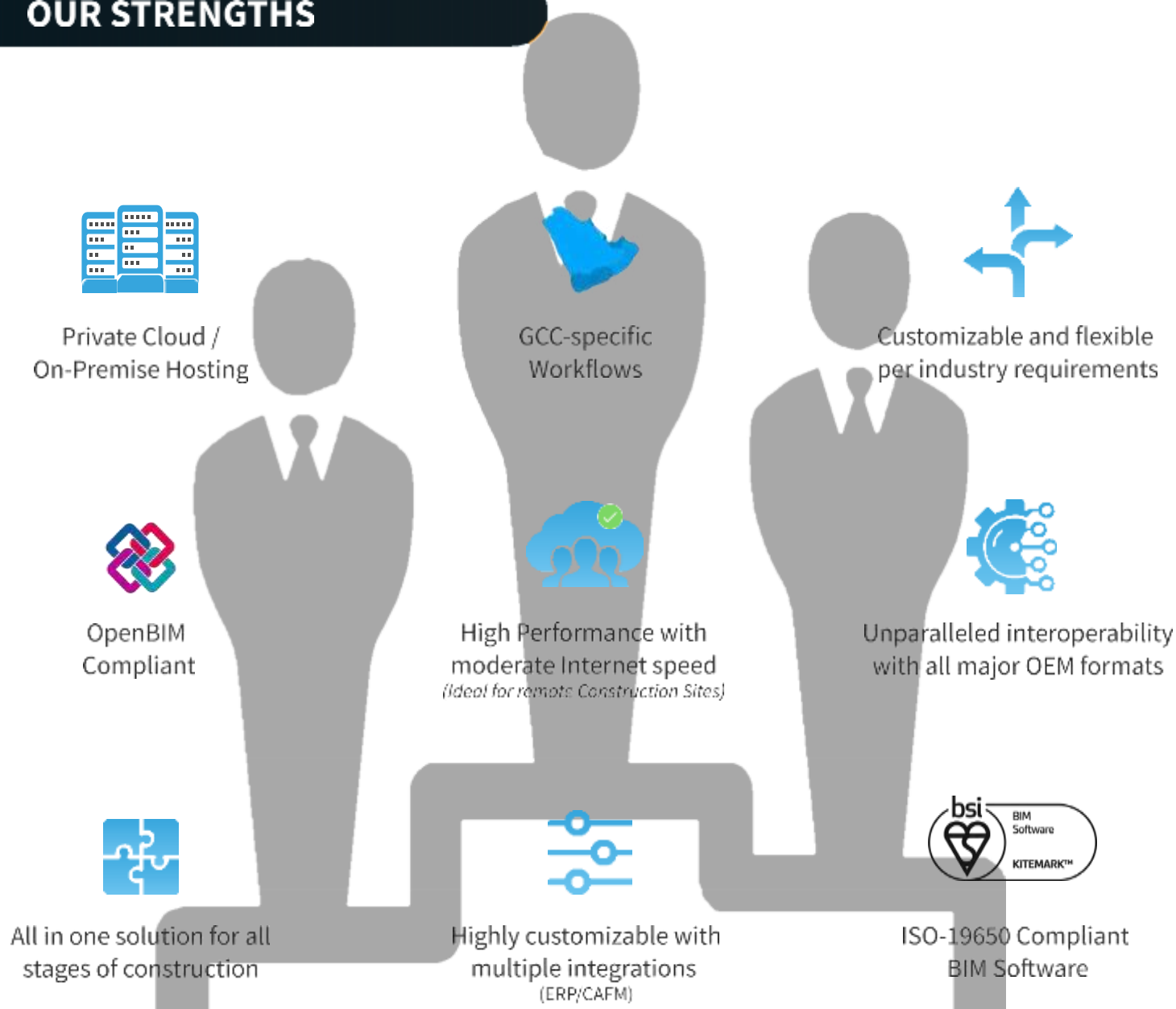
✓ ENERGY SAVINGS

With performance data and analysis, operators can optimize energy consumption

31%

As a result of COVID-19, 31% of respondents use digital twins to improve employee or customer safety, such as the use of remote asset monitoring to reduce the frequency of in-person monitoring. - Gartner.

OUR STRENGTHS



“ Unlike other platforms, our philosophy at CUBE is to adapt to the customer workflows rather than forcing them to implement pre-defined templates ”

OUR PRESENCE



GOVT. OF INDIA E-CONSTRUCTION PORTAL

Client:



In association with:



Problem Statement

- CPWD wanted to digitalize their existing file viewing, reviewing, sharing and approval processes
- Moreover, being one of the critical government organizations in India, they also wanted to store their data securely within the country and within their own servers
- The existing common data environment solutions and other collaboration platforms available, were not able to cater to this requirement. On top of this, the tools were limited to customizability and came with a huge price.

Our Solution

- CUBE was a custom solution deployed for CPWD along with Tech Mahindra and customized in line to cater to CPWDs requirements
- An advanced document management module was developed for storing and viewing files, along with a powerful & robust 2D - CAD and 3D - BIM viewer to view and navigate through the 2D drawings and 3D BIM models
- Issue management module was developed through which the CPWD officials could mark issues and review documents and share it with the relevant stakeholders and project owners
- We also developed a Quantity and Cost module, which helped the authorities to extract precise quantities of materials from both 2D & 3D files and link them to the respective costs as per DSR
- We further customized the document approval workflows as well as the user management and permission matrix as per CPWD organization standards for their ease and convenience
- The Collaboration portal and the data within it was stored on cloud - in secure servers procured by the government of India

Modules & Features Implemented:



CAD & BIM
Viewer



Hosted on
Private Cloud



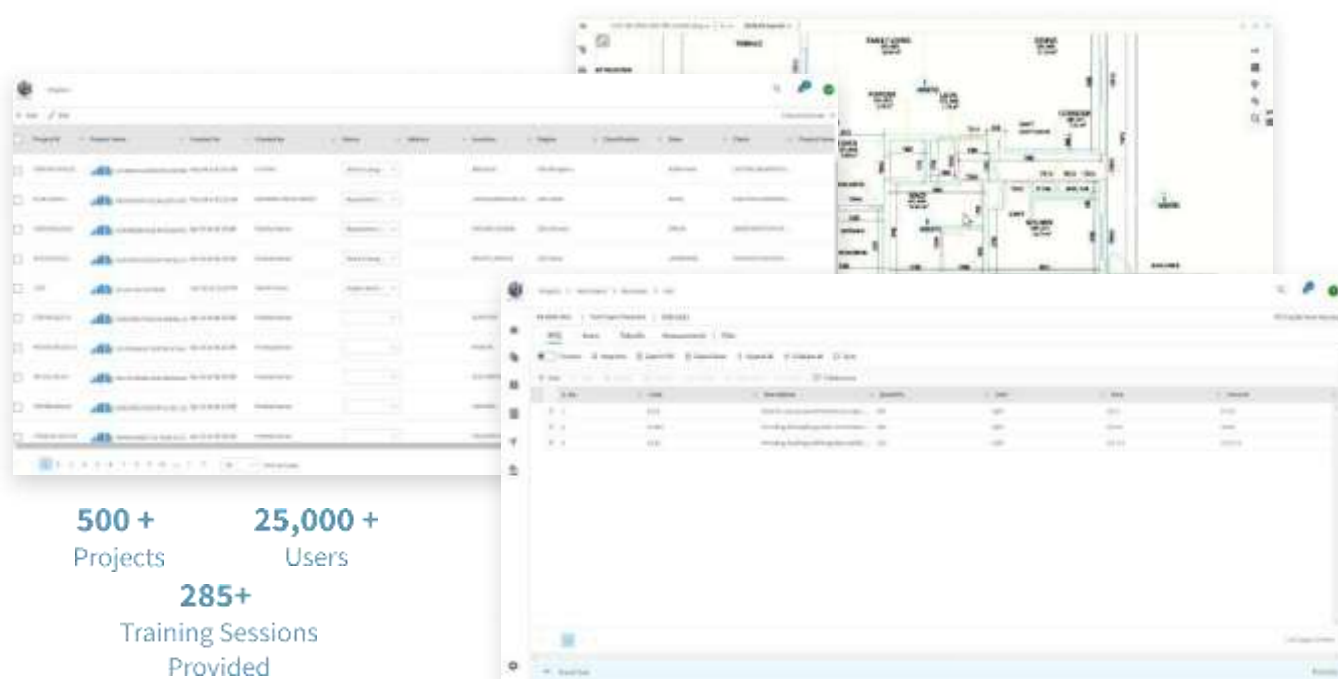
Document
Management



Estimations
& Costs



Issue
Management



500 +
Projects

25,000 +
Users

285+
Training Sessions
Provided

MADHYA PRADESH METRO

Client:



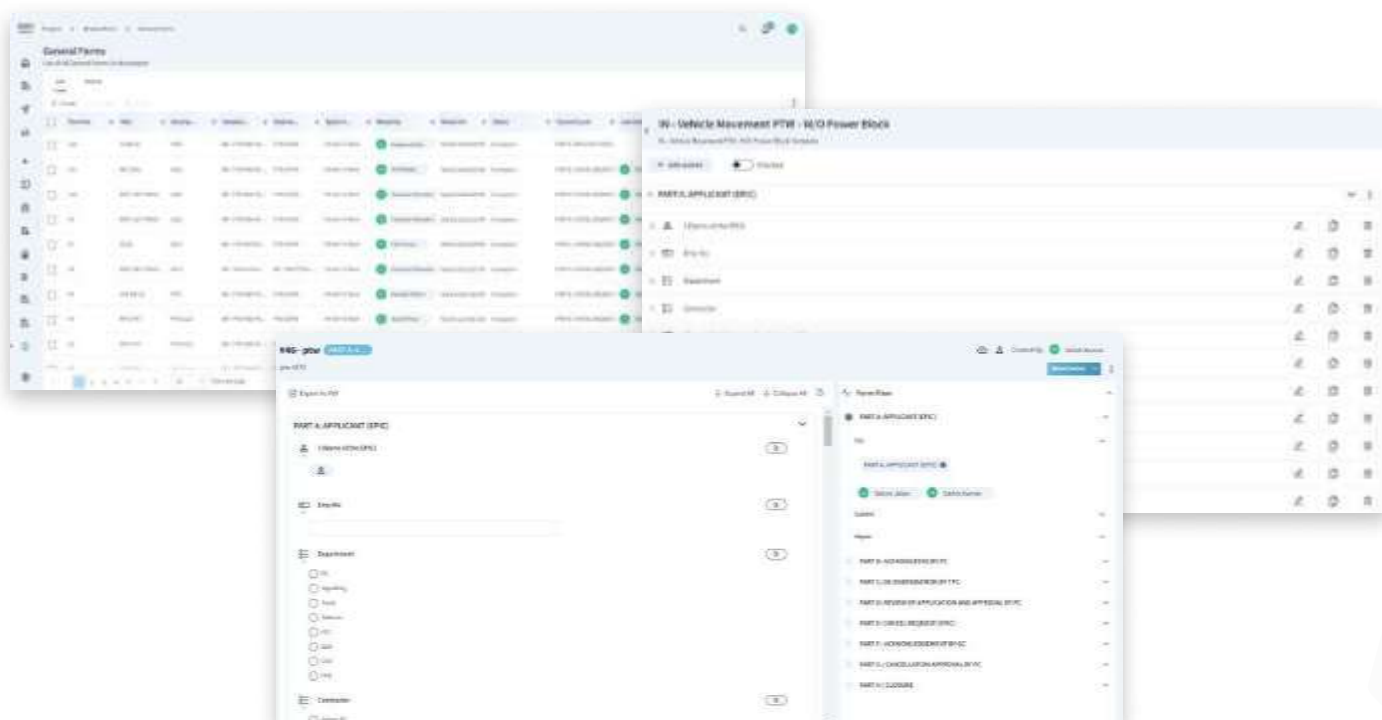
MPMETRO
Rail Corporation Limited

Problem Statement

- Madhya Pradesh Metro Rail Corporation Limited (MPMRCL) required a cloud-based solution for managing Design, Supply, Implementation, Support and Maintenance activities using a Project Management Information System for Indore Metro Rail Project and Bhopal Metro Rail Project
- MPMRCL required a solution for Contracts Management, Project planning, a viewer to view CAD and BIM files, issue management, task management, document management, Project reporting and monitoring, and remote data capture

Our Solution

- Customized platform to meet specific requirements of MPMRCL
- Developed advanced document management module with an organizational folder structure
- Linked schedules and estimates for efficient subcontractor scope allocation
- Implemented an issue management module for quick problem resolution and improved stakeholder management
- Configured daily progress reports, inspection checklists, and request forms for enhanced site correspondence
- Tailored document approval workflows and user responsibility matrix
- Introduced asset management module for updating metadata and creating digital twins for effective facility management
- Stored Collaboration portal data securely on MeitY empanelled cloud servers



400 BEDDED HOSPITAL, UAE

Problem Statement:

- The client faced continuous challenges in managing their operations and maintenance data of numerous assets in the Hospital Building
- They wanted an automated digital system to manage and monitor their live asset, accessible through cloud

Our Solution:

- The digital twin management feature of CUBE was used to overcome this challenge
- The work orders, specification manuals, as-built drawings and other asset-related documents were linked to the assets within the BIM model and stored in a single platform on the cloud
- This ensured that the data was stored securely on the cloud, and facility managers could access this data at their convenience
- The IoT sensors installed on-site were linked to the BIM model through the CUBE platform
- This ensured that the assets were monitored 24/7, and in case of any malfunctions to these assets, they were intimated to the facility managers using an intelligent alert system. Color codes were used to automatically identify assets near the preventive maintenance date.



Modules & Features Implemented:



CAD & BIM
Viewer



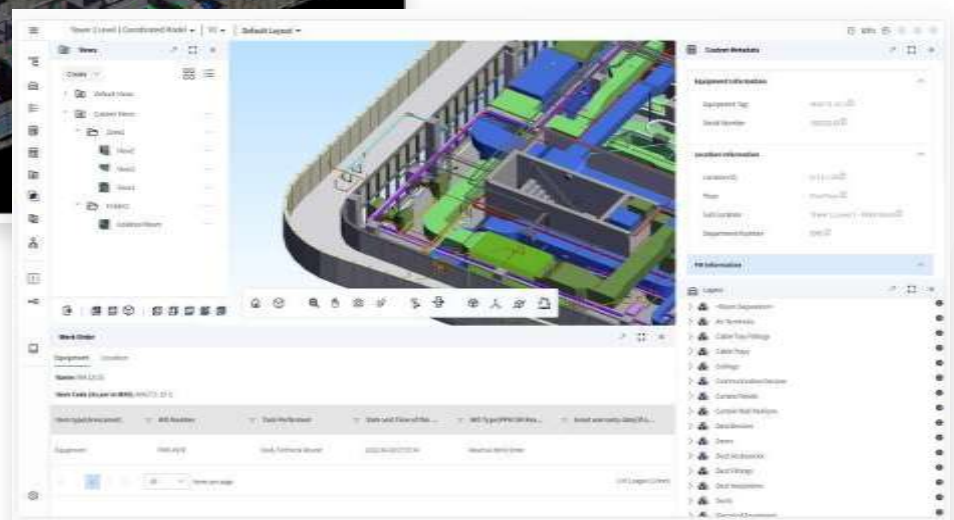
Document
Management



Asset & Digital
Twin Platform



Issue
Management



GENERAL CONTRACTOR BASED IN USA

Problem Statement:

- The client worked in the AEC industry and extensively used BIM and related technologies to execute their projects.
- The existing CDE used by them could not be customized as per their organization standards, nor were they economically feasible.
- They were exploring and looking out for a CDE that could be customizable as per their workflows, affordable as well as include all the advanced features and modules necessary for a design and build firm.



Our solution:

- The client reached out to us looking out for a solution that could do the existing functionalities of their CDE but also be customized according to their workflows as well as fit within their budget.
- The document management module was implemented along with our CAD & BIM viewer and the Issue management functionality. A custom portal was created with their company logo and a custom URL, and tailored the system as per their workflows.
- The client used the document management module to store all the files related to the project and also viewed the drawings and BIM models on the cloud.
- They further marked design & coordination issues through the viewers and also tracked them through the issue tracking functionality.

They also sent the issues as a transmittal to the respective design consultants and made sure they were notified of the design discrepancies.



Modules & Features Implemented:



CAD & BIM
Viewer



Issue
Management



Document
Management

CUBE PORTFOLIO



Gate Avenue Mall DIFC, Dubai



ESCR Parks New York



Community Park Building Dubai



CPWD Projects



Bhopal Metro



Indore Metro



Maharashtra Jeevan Pradhikaran



IRCON - Haridwar ring road project



Vijaywada Airport



Rustomjee Projects



River Ganga Bridge (2.5KM)



Ranchi Railway Station



Trichy Airport



Ajni Railway Station



Kanpur Central Railway Station

